

By Georgina Rannard, Becky Dale and Erwan Rivault
BBC News Climate & Science and Data Journalism Team

The sea-ice surrounding Antarctica is well below any previous recorded winter level, satellite data shows, a worrying new benchmark for a region that once seemed resistant to global warming.

Antarctica's huge ice expanse regulates the planet's temperature, as the white surface reflects the Sun's energy back into the atmosphere and also cools the water beneath and near it.

Without its ice cooling the planet, Antarctica could transform from Earth's refrigerator to a radiator, experts say.

The ice that floats on the Antarctic Ocean's surface now measures less than 17 million sq km - that is 1.5 million sq km of sea-ice less than the September average, and well below previous winter record lows.

Scientists are still trying to identify all the factors that led to this year's low sea-ice - but studying trends in Antarctica has historically been challenging.



Very thin sea-ice in the foreground - this is a type of sea-ice called "nilas" that forms in very low wind conditions

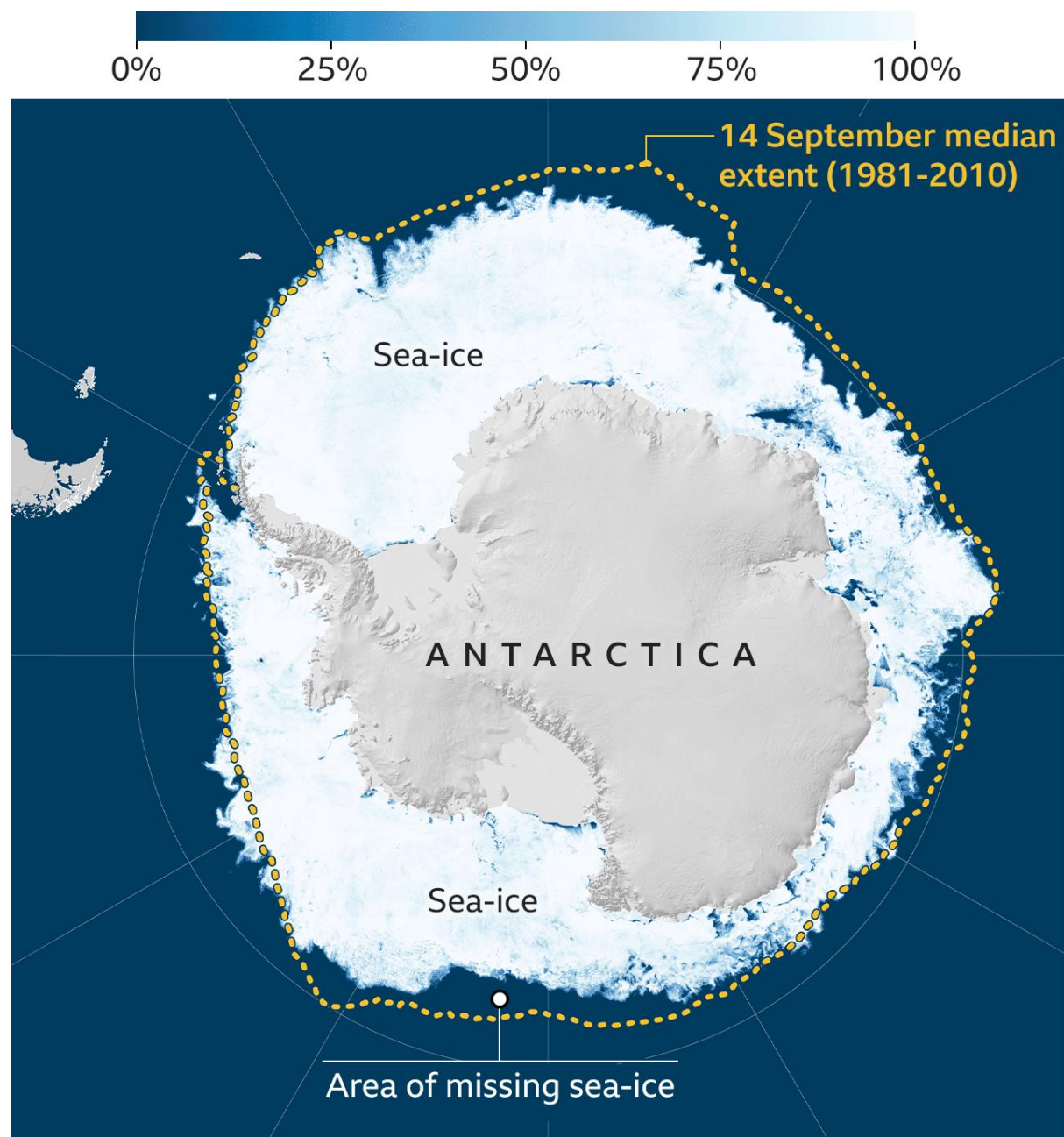
Sea-ice forms in the continent's winter (March to October) before largely melting in summer, and is part of an interconnected system that also consists of icebergs, land ice and huge ice shelves - floating extensions of land ice jutting out from the coast.

Sea-ice acts as a protective sleeve for the ice covering the land and prevents the ocean from heating up. As more sea-ice disappears, it exposes dark areas of ocean, which absorb sunlight instead of reflecting it, meaning that the heat energy is added into the water, which in turn melts more ice. Scientists call this the ice-albedo effect.

That could add a lot more heat to the planet, disrupting Antarctica's usual role as a regulator of global temperatures.

Winter sea-ice is missing in some areas

Sea-ice concentration, 14 September 2023



Source: National Snow and Ice Data Center (NSIDC), Polar Bremen

B B C

Since the 1990s, [the loss of land ice from Antarctica has contributed 7.2mm](#) to sea-level rise.

Even modest increases in sea levels can result in dangerously high storm surges that could wipe out coastal communities. If significant amounts of land ice were to start melting, the impacts would be catastrophic for millions of people around the world.

Antarctica's remoteness and shortage of historical information means a lot is still unknown.

Scientists know how far the sea-ice spreads, but not, for instance, how thick it is. Unlocking that puzzle could radically change climate models for the region.

At the scientific base Rothera, Dr Mallet is using radar instruments to study sea-ice thickness for an international research project called Defiant.

"There is a chance that it's a really freak expression of natural variability," he says, meaning that lots of natural factors could have built up and are affecting the region simultaneously.

This year's [record-warm oceans](#) are likely a contributing factor, scientists suggest - warm water will not freeze.

And there may have also been changes in ocean currents and the winds that drive temperatures in the Antarctic.